Mandatory APS ELN Practice Assignment

Deadline: November 16, 2026
Online Textbook: http://10.109.92.41/MasteringELN/ | **PDF Book:** http://10.109.92.41/MasteringELN/Mastering-ELN.pdf
Assignment Page: http://10.109.92.41/MasteringELN/assignment/
Every student must complete both practice records inside their personal ELN account by the deadline. Both projects will be audited automatically on the laboratory server after November 16.

Step 1: Download the Raw Data Packages from the Server

Download both raw instrument bundles from the CSMIC server and unpack them into two separate folders on your computer:

1. **Study A (`261015A`, BN/DLC tribological coating, 11 raw files):**
   http://10.109.92.41/MasteringELN/data/Study-A-261015A-raw.zip
   Contains `.depo.json`, `.run.json`, `.tsv`, three step-height `.fibps` scans (`1.fibps`, `2.fibps`, `3.fibps`), nanoindentation (`BN-DLC(261015A).mit`), reciprocating friction test (`BN-DLC.zrw`), three wear-track `.fibps` scans (`wear 1.fibps`, `wear 2.fibps`, `wear 3.fibps`), and `checksums.md5`.
2. **Study B (`261014B`, Co/C multilayer X-ray mirror, 7 raw files):**
   http://10.109.92.41/MasteringELN/data/Study-B-261014B-raw.zip
   Contains the 3 deposition files, the 3 plasma-cleaning files, the raw X-ray reflectivity scan `XRR-1_Co-C(261014B).xrdml`, and `checksums.md5`. The `.xrcx` fit file is intentionally omitted because you will produce it by fitting the `.xrdml` curve in Task 2.

Step 2: Create Your Personal Practice Group and Two Practice Projects

Log into **APS ELN** (`ELabNotebook3.exe`) with your own laboratory account. Do not record practice data inside shared production projects:

1. In the left **Projects** panel, choose **Project > New Group** (or right-click an empty area of the Projects panel and choose **New Group...**). Enter the exact group name `Practice` and click **OK** (*Chapter 5: Creating, sorting, and moving projects and folders*).
2. Right-click your new `Practice` group and choose **New Project...**. Create your first practice project with the exact title `Study A Practice` and click **Save**.
3. Right-click your `Practice` group a second time and choose **New Project...**. Create your second practice project with the exact title `Study B Practice` and click **Save**.

Remember: In the Commercial team edition, project groups are managed in the Desktop window. Creating `Study B Practice` inside your `Practice` group before launching your AI agent ensures both practice projects sit cleanly inside `Practice` under your user node.

Step 3: Complete Task 1 and Task 2

Task 1: Study A (`261015A`) in the Desktop GUI (inside `Study A Practice`)

Read **Chapters 1–4** of *Mastering ELN* (http://10.109.92.41/MasteringELN/ch4-your-first-complete-sample-record/) and follow the four steps of **Chapter 4** inside your `Study A Practice` project:

1. **Build the 6-item tree first:** `Deposition 261015A` (Procedure, Action type `Deposition`) → `261015A` (Specimen) → child actions `Thickness 261015A` (`3D Profiling`), `Indentation 261015A` (`Indentation`), and `Friction test 261015A` (`Tribotest`), plus child action `Wear track 261015A` (`3D Profiling`) under `Friction test 261015A`.
2. **Attach all 11 raw files** from `Study-A-261015A-raw.zip` to their matching items (*Table 4-1*).
3. **Extract all 23 `Reader` values** using **Extract Values** and **Extract values from 3 files...**:
   - 12 values on `Deposition 261015A` from `.depo.json` (10 conditions + 2 results, `aps.deposition` v1);
   - 3 step-height values on `Thickness 261015A` from `1.fibps`, `2.fibps`, `3.fibps` (`aps.fibps` v3);

- 2 values (`Hardness` and `Modulus` with STD tolerances) on `Indentation 261015A` from `BN-DLC(261015A).mit` (`aps.mit`);
- 3 values (`Fn`, `Cycles`, and `COF`) on `Friction test 261015A` from `BN-DLC.zrw` (`aps.zrw` v2);
- 3 wear-area values on `Wear track 261015A` from `wear 1.fibps`, `wear 2.fibps`, `wear 3.fibps` (`aps.fibps` v3).

4. **Compute the 1 `Derived` wear rate** on `Friction test 261015A` by right-clicking `Friction test 261015A` and choosing **Add > Derived Value...** (`wear_rate_reciprocating`, K = 3.05 × 10^{-7} $mm^3/(N \cdot mm)$), selecting all three `Wear Area` values plus `Fn` and `Cycles`. Then right-click `Deposition 261015A` and run **Check Value Sources...** to confirm all 24 values report `unchanged`.

Task 2: Study B (`261014B`) with Your AI Agent and `XRC_MCP` (inside `Study B Practice`)

Read **Chapters 17–19** of *Mastering ELN* (http://10.109.92.41/MasteringELN/ch19-study-b-xrr-fitting-with-provenance/). Copy the 7 raw files from `Study-B-261014B-raw.zip` into your configured `--upload-root` directory and use your AI agent (`ELNMCPServer.exe` + `XRC_MCP.exe`) inside your `Study B Practice` project:

1. **Build the tree and extract 18 deposition/cleaning values (*Chapter 19, Part 1*):** Have your agent create folder `261014B` inside `Study B Practice`, create the 4 items (`Deposition 261014B` [Proc], `Cleaning 261014B` [Action], `261014B` [Specimen], and `XRR-1 261014B` [Action]), upload the 7 raw files via `upload_file`, and extract and store the 18 `Reader` values (14 on `Deposition 261014B` and 4 on `Cleaning 261014B`) from the two `.depo.json` files via `extract_file_key_values` and `add_values`.
2. **Fit `XRR-1_Co-C(261014B).xrdml` via `XRC_MCP` (*Chapter 19, Part 2*):** Have your agent assess and fit the reflectivity scan in `XRC_MCP.exe` (`SodaLimeGlass` substrate / [C / Co] × 20 main stack / surface C contamination layer, polynomial order 1) following the laboratory 13-step fitting procedure (see http://10.109.92.41/MasteringXRR/). Save the accepted `.xrcx` project using `curves.job_id` (without a manually typed `structure` override) so `[FITRESULT]` and χ^2 are written into the archive.
3. **Upload the `.xrcx` fit and extract the 4 fit values:** Have your agent create child action `XRR fit 261014B` under `XRR-1 261014B`, upload the saved `.xrcx` file, and extract and store the 4 `Reader` values (`Period`, `Co Thickness`, `C Thickness`, and `Chi2`) via `aps.xrcx`.
4. **Run the three MCP health checks:** Have your agent run `check_storage` (8 ok), `check_value_drift` (22 checked, 22 unchanged), and `report_value_issues` (0 issues) on `Study B Practice`.

> **Warning (No Manual Values):** Every stored number in both projects is checked for `Reader` or `Derived` provenance. Typing a number by hand (`Manual` source), extracting from `.tsv` logs instead of `.depo.json`, or saving an `.xrcx` without `curves.job_id` (which writes `[LASTFIT]` with no `Chi2`) fails the automated audit.

Step 4: Pre-Submission Checklist Before November 16, 2026 (Chapter 23)

Before **November 16, 2026**, verify both projects in your `Practice` group against the ten checks in **Chapter 23** (http://10.109.92.41/MasteringELN/ch23-ten-checks-before-you-sign-or-submit/):

Project in `Practice` Group	Items	Attached Files	Stored Values	Required Checks Before Nov 16
`Study A Practice` (Sample `261015A`, GUI)	6 items	11 raw files	23 `Reader` + 1 `Derived` (24 total, 0 `Manual`)	**Data Quality Check...** (0 issues) & **Check Value Sources...** (24 unchanged)
`Study B Practice` (Sample `261014B`, MCP)	5 items	7 raw files + 1 fitted `.xrcx` (8 total)	22 `Reader` (14 depo + 4 clean + 4 XRR fit, 0 `Manual`)	`check_storage` (8 ok), `check_value_drift` (22 unchanged), `report_value_issues` (0 issues)